

2023 Ph H2 Q6

Section: Our Dynamic Universe

Topic: The Expanding Universe

Summary of Question:

The lyrics of the Big Bang Theory theme song are quoted. Using knowledge of physics, comment on the lyrics.

Comment on lyrics

“Hot, dense state” → describes Big Bang.

“14 billion years ago” → current estimate $\approx$ 13.8 billion years.

“Elements were made in a fraction of a second” → hydrogen & helium formed within first 3 minutes; heavier elements formed later in stars.

“Expanding outward” → consistent with Hubble’s redshift evidence.

“Stars go the other way” → implies Big Crunch. Current evidence suggests expansion is accelerating (dark energy), so contraction unlikely.

Answer: Broadly correct, except for heavy element formation and prediction of collapse.

Final Answers

Universe began $\sim$ 13.8 billion years ago.

Expansion continues today (accelerating).

Light elements formed early; heavier ones in stars.

Big Crunch not supported by evidence.

Revision Tips

- Universe age ~13.8 billion years (CMB evidence).
- Nucleosynthesis produced light elements early.
- Hubble's law shows galaxies receding.
- Expansion is accelerating due to dark energy.